

## Scatter Plots and Bivariate Association — Association vs. Cause Answer Key

Grade 8 / Pre-Algebra

### Quick Answers

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|       |       |       |         |
|-------|-------|-------|---------|
| 1. 12 | 3. —  | 5. 73 | 7. -2.5 |
| 2. 14 | 4. 84 | 6. —  | 8. —    |

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### Curriculum

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**Level:** Grade 8 / Pre-Algebra

6.SP.B.5 – problems 4, 5

7.SP.A – problem 6

8.SP.A.1 – problems 1, 2, 3, 7, 8

*Difficulty 1–5 of 5 across 8 tagged checks.*

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### Common wrong answers

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2. *If they got 28: subtracted the ice-cream numbers on the horizontal axis instead of the sunscreen numbers being asked about*

4. *If they got 78.5: averaged everyone in the study together instead of the app group alone*

7. *If they got 5: subtracted the television hours on the horizontal axis instead of the sleep hours being asked about*

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### 1. Graph A. In the week when 26 cartons were sold, how many bottles of sunscreen were sold?

Step 1: find 26 on the horizontal axis — that axis carries the ice-cream counts, which is the quantity the question gives.

Step 2: go straight up to the plotted point, then straight across to the vertical axis, which carries the sunscreen counts.

Step 3: the point sits at height 12:

|    |
|----|
| 12 |
|----|

The ordered pair is (26, 12): 26 cartons, 12 bottles.

**2. Graph A. Dana compared the 10-carton week with the 38-carton week and wrote “sunscreen went up by 28”. How many more bottles of sunscreen were really sold?**

Step 1: name Dana’s mistake —  $38 - 10 = 28$  is the change in *cartons*, read off the horizontal axis. The question asks about sunscreen, which lives on the vertical axis.

Step 2: read both sunscreen values. The 10-carton week has 4 bottles; the 38-carton week has 18 bottles.

Step 3: subtract the two *vertical* values:  $18 - 4$ :

14

Step 4: sanity check — sunscreen sales cannot rise by more than the highest value plotted on that axis, so 28 was impossible from the start.

**3. Graph A. Dana claims the plot proves ice-cream sales cause sunscreen sales. Name a lurking variable and say why the plot cannot show cause. (Open response — review by hand.)**

*Model answer.* The lurking variable is the **weather** — the temperature, or how sunny the day is. Hot, sunny weeks send people out for both ice cream and sunscreen, so both counts rise together without either one causing the other. Graph A only records what happened alongside what; nobody changed the ice-cream sales and then watched sunscreen respond. An association is a *pattern in the pairs*, and a pattern is consistent with three different stories: ice cream drives sunscreen, sunscreen drives ice cream, or a third variable drives both.

*Look for:* a named third variable that plausibly moves **both** quantities (heat, sunshine, summer holidays, tourist season — not “people like summer”), plus the idea that association alone does not pick between the possible stories.

**4. Table B. Ravi averaged all ten scores and got 78.5. Find the correct mean for the group that used the app.**

Step 1: name Ravi’s mistake — he pooled both groups. A group mean uses only that group’s data; mixing in the non-users pulls the app group’s mean down toward the middle, which is exactly what 78.5 is.

Step 2: add the five app scores:  $88 + 79 + 95 + 84 + 74 = 420$ .

Step 3: divide by how many scores there are, which is five:  $420 \div 5$ :

84

Step 4: check the answer is believable — 84 sits inside the app group’s range from 74 to 95 ✓, while 78.5 was being dragged down by scores that do not belong to this group.

**5. Table B. Find the mean score of the group that did not use the app.**

Step 1: use only the second row of the table — the same rule as problem 4, applied to the other group.

Step 2: add:  $72 + 66 + 81 + 70 + 76 = 365$ .

Step 3: divide by five:  $365 \div 5$ :

73

Step 4: compare the two group means: 84 against 73, a gap of 11 points in favour of the app users. The gap is real — what problem 6 asks is what the gap is *evidence of*.

**6. Table B. The advertisement claims the app raises scores by eleven points. Why can this study not support that, and what would?** (*Open response — review by hand.*)

*Model answer.* The students **chose for themselves** whether to use the app, so the two groups were not alike before the app was introduced. The kind of student who downloads a tutoring app is often already the student who studies more, sleeps enough, has a quiet place to work, or started the year ahead. Any of those differences could produce an eleven-point gap on its own, so the gap cannot be assigned to the app. This is a self-selected observational study, and it can establish association only.

*The fix: random assignment.* Take one group of students, and by coin flip (or a random-number list) assign each one to use the app or not. Random assignment makes the two groups alike on average in every respect — including things nobody thought to measure — so a score gap that shows up afterwards can be credited to the app. Keeping everything else the same for both groups, and using enough students, is part of the same fix.

*Look for:* the words self-selected / not randomly assigned / the groups may already differ, at least one concrete pre-existing difference, and random assignment (or a randomized controlled experiment) as the remedy.

**7. Graph C. Sam reported the effect as 5 by subtracting television hours. Going from the 1-hour student to the 6-hour student, what is the change in hours of sleep?**

Step 1: name Sam’s mistake —  $6 - 1 = 5$  is the change in *television* hours, the horizontal axis. Sleep is the vertical axis.

Step 2: read the two sleep values: the 1-hour student sleeps 9 hours, the 6-hour student sleeps 6.5 hours.

Step 3: change means end value minus start value:  $6.5 - 9$ :

|      |
|------|
| -2.5 |
|------|

Step 4: read the sign — it is negative, so sleep *falls* by 2.5 hours across that range. That is a negative association, which is all the graph reports.

**8. Challenge — Graph C. Rewrite Sam’s headline as a claim the graph supports, then describe the experiment that would test the cause claim.** (*Open response — review by hand.*)

*Model rewrite.* “Among these six students, more television hours went with fewer hours of sleep.” Other acceptable forms: “Television time and sleep are negatively associated in this sample,” or “Students who watched more television slept less on this night.” What must go: *makes, causes, leads to*, and any claim about students in general.

*Model experiment.* Recruit a large group of similar students. Assign each student at random — coin flip or random-number list, not by preference — to a fixed television allowance for the evening: say a low-television group and a high-television group. Hold bedtime routine, homework load and wake-up time the same for everyone, measure each student’s sleep, and repeat over several nights. Because the television time was assigned rather than chosen, a difference in sleep between the groups can be credited to television.

*Look for:* a rewritten claim with no causal verb and scoped to the data collected; and an experiment with (a) random assignment, (b) an assigned television level rather than a self-reported one, and (c) other conditions held the same. A plan that merely surveys more students still cannot show cause — that is the point of the whole sheet.